Chapter 8
Operational Amplifier as A Black Box

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Chapter 8 Operational Amplifier as A Black Box

- 8.1 General Considerations
- 8.2 Op-Amp-Based Circuits
- 8.3 Nonlinear Functions
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- 8.5 Design Examples

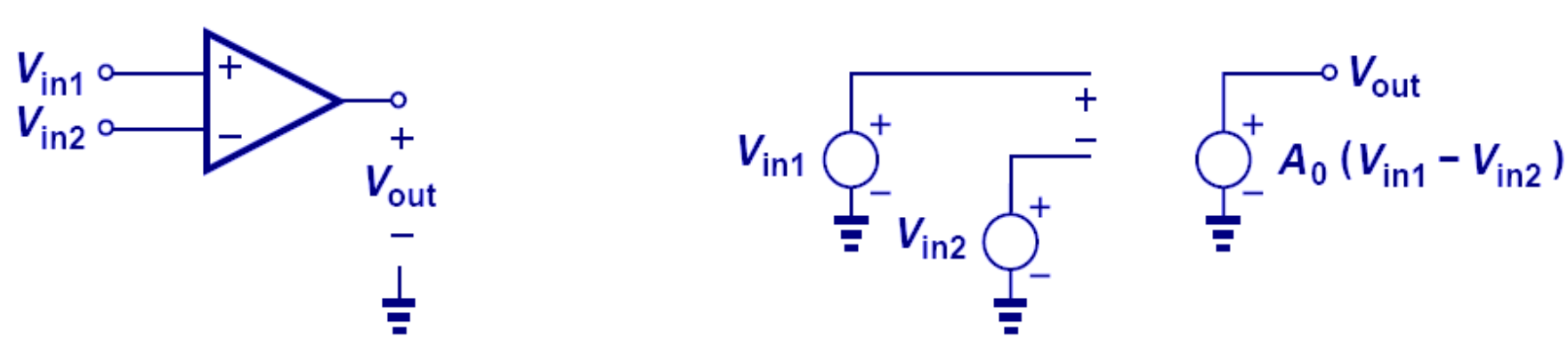
Outline

- The outline below illustrates the sequence of concepts covered in this chapter.

General Concepts	Linear Op Amp Circuits	Nonlinear Op Amp Circuits	Op Amp Nonidealities
• Op Amp Properties	• Noninverting Amplfier • Inverting Amplifier • Inetgrator and Differentiator • Voltage Added	• Precision Rectifier • Logarithmic Amplifier • Square Root Circuit	• DC Offsets • Input Bias Currents • Speed Limitations • Finite Input and Output Impedances

Basic Op Amp

- Op amp is a circuit that has two inputs and one output.
- It amplifies the difference between the two inputs.

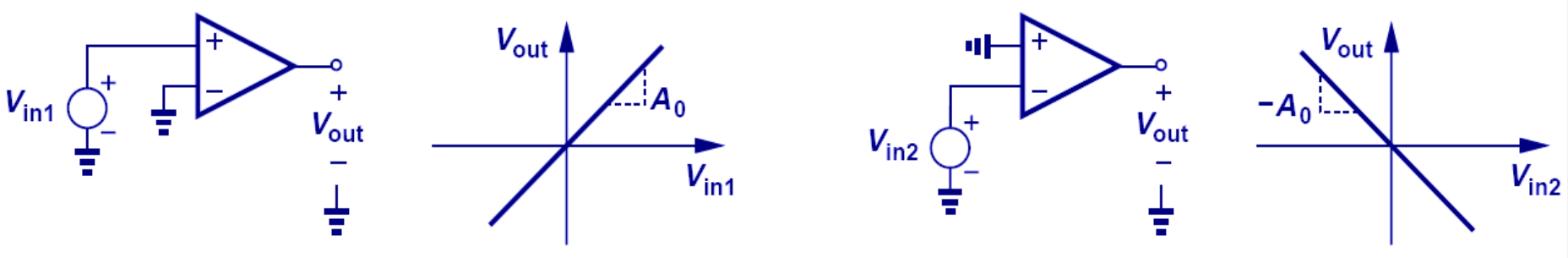


$$V_{out} = A_0(V_{in1} - V_{in2})$$

where A_0 is the "open-loop" gain

Inverting and Non-inverting Op Amp

- If the negative input is grounded, the gain is positive.
- If the positive input is grounded, the gain is negative.



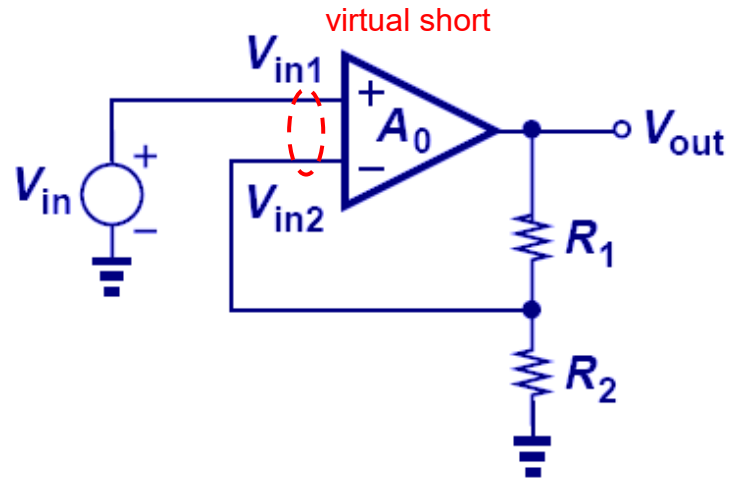
Ideal Op Amp

- Infinite gain ($A_0 = \infty$)
- Infinite input impedance ($R_{in} = \infty$)
- Zero output impedance ($R_{out} = 0$)
- Infinite speed ($\mathrm{BW} = \infty$)

Virtual Short ($V_+ \approx V_-$)

- Virtual short: $V_{in} \to 0$ ($V_{in1} \approx V_{in2}$)

  Due to infinite gain of op amp, the circuit forces V_{in2} to be close to V_{in1}, thus creating a virtual short.

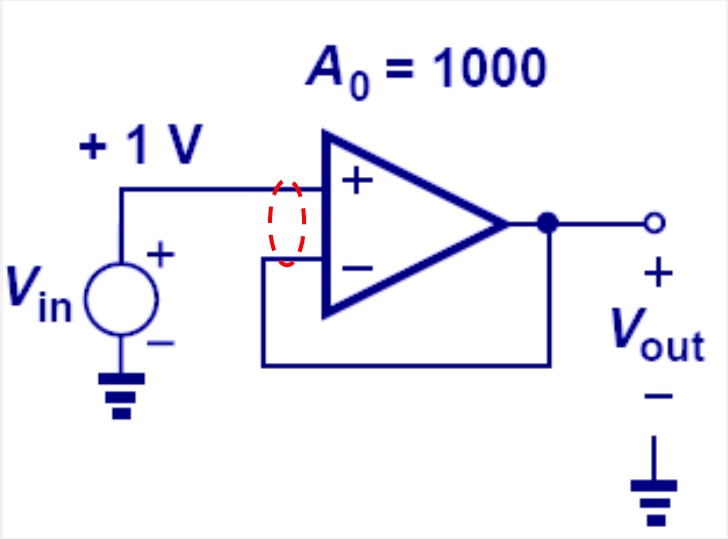


For an ideal op amp, $A_0 = \infty \Rightarrow V_{out}$ is finite

$$A_0 = \frac{V_{out}}{V_{in}} \Rightarrow \infty = \frac{V_{out}}{0}$$

$$V_{in} \to 0 \ (V_{in1} \approx V_{in2})$$

Unity Gain Amplifier



Method #1: Without using the virtual short concept (finite A_0)

$$V_{out} = A_0(V_+ - V_-) = A_0(V_{in} - V_{out})$$

$$\frac{V_{out}}{V_{in}} = \frac{A_0}{1 + A_0} = \frac{1000}{1 + 1000} \approx 1$$

Method #2: Apply the virtual short concept (infinite A_0),

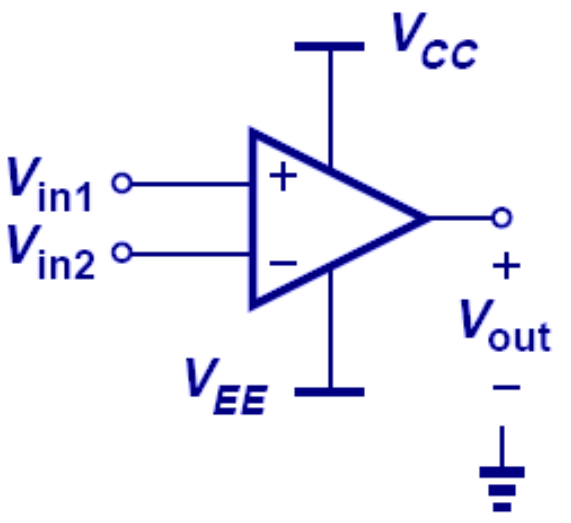
$$V_+ = V_-$$

$$V_{in} = V_{out}$$

$$A_v = \frac{V_{out}}{V_{in}} = 1$$

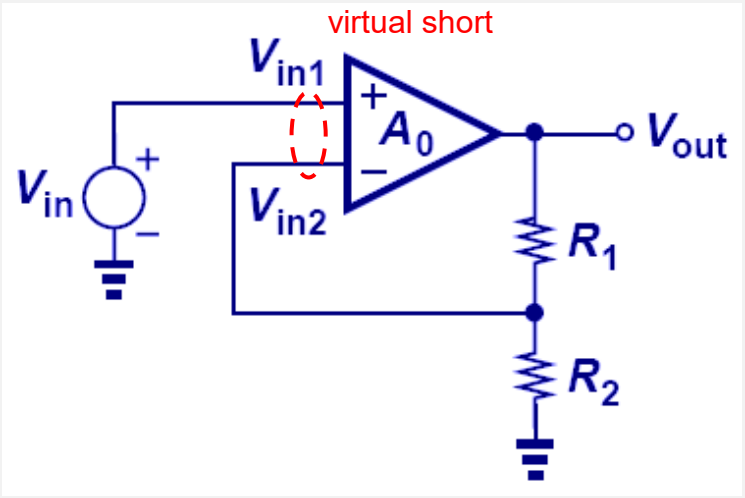
Op Amp with Supply Rails

- Dual power supply (bipolar): to explicitly show the supply voltages, V_{CC} and V_{EE} are shown.
- Single power supply (unipolar): V_{EE} is zero.



Noninverting Amplifier (Infinite A_0)

- A noninverting amplifier returns a fraction of output signal thru a resistor divider to the negative input.
- With a high A_o, V_{out}/V_{in} depends only on ratio of resistors, which is very precise.

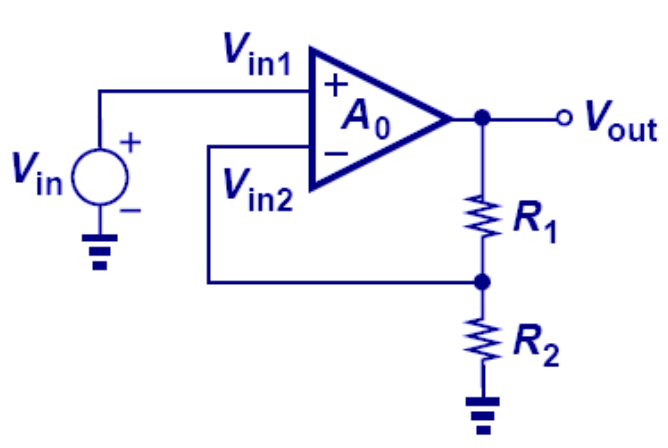


Apply the virtual short concept, $V_{in1} = V_{in2}$

$$V_{in} = V_{in1} = V_{in2} = \frac{R_1}{R_1 + R_2} V_{out}$$

$$A_v = \frac{V_{out}}{V_{in}} = \frac{R_1 + R_2}{R_1} = 1 + \frac{R_1}{R_2}$$

Noninverting Amplifier (Finite A_0)



$$V_{out} = A_0(V_{in1} - V_{in2}) = A_0\left(V_{in} - \frac{R_2}{R_1 + R_2}V_{out}\right)$$

$$A_v = \frac{V_{out}}{V_{in}} = \frac{A_0}{1 + \frac{R_2}{R_1 + R_2}A_0}$$

If $\frac{R_2}{R_1 + R_2}A_0 \gg 1$, $\quad A_v \approx \frac{A_0}{\frac{R_2}{R_1 + R_2}A_0} \approx \frac{R_1 + R_2}{R_1} = 1 + \frac{R_1}{R_2}$

Rewrite A_v as follows,

$$A_v = \frac{1 + \frac{R_1}{R_2}}{1 + \underbrace{\left(1 + \frac{R_1}{R_2}\right)\frac{1}{A_0}}_{\triangleq \varepsilon}}$$

For $\varepsilon \ll 1$, $(1 + \varepsilon)^{-1} \approx 1 - \varepsilon$.

$$A_v \approx \left(1 + \frac{R_1}{R_2}\right)\left[1 - \left(1 + \frac{R_1}{R_2}\right)\frac{1}{A_0}\right],$$

where $\left(1 + \frac{R_1}{R_2}\right)\frac{1}{A_0}$ is the "gain error," it must be minimized according to each application's requirements. The larger the closed loop gain, the more inaccurate the circuit is.

Example 8-3

- A noninverting amplifier incorporates an op amp having a gain of 1000. Determine the gain error if the circuit is to provide a nominal gain of (a) 5, or (b) 50.

Solution:

(a)

For a nominal gain of 5, we have

$$\left(1+\frac{R_1}{R_2}\right)=5.$$

The gain error is

$$\left(1+\frac{R_1}{R_2}\right)\frac{1}{A_0}=(5)\frac{1}{1000}=0.5\%$$

(b)

For a nominal gain of 50, we have

$$\left(1+\frac{R_1}{R_2}\right)=50.$$

The gain error is

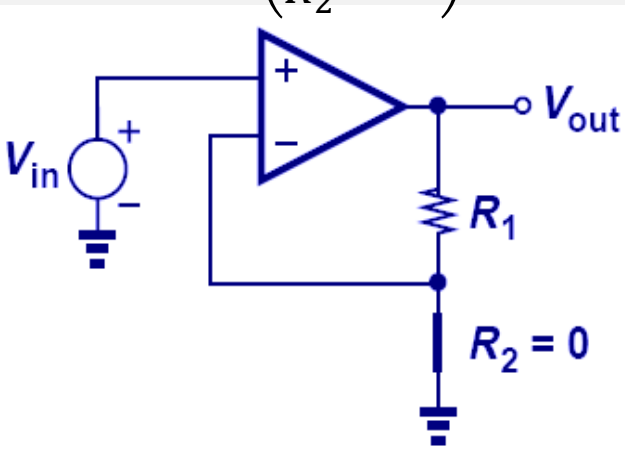
$$\left(1+\frac{R_1}{R_2}\right)\frac{1}{A_0}=(50)\frac{1}{1000}=5\%$$

In other words, a higher closed-loop gain inevitably suffers from less accuracy.

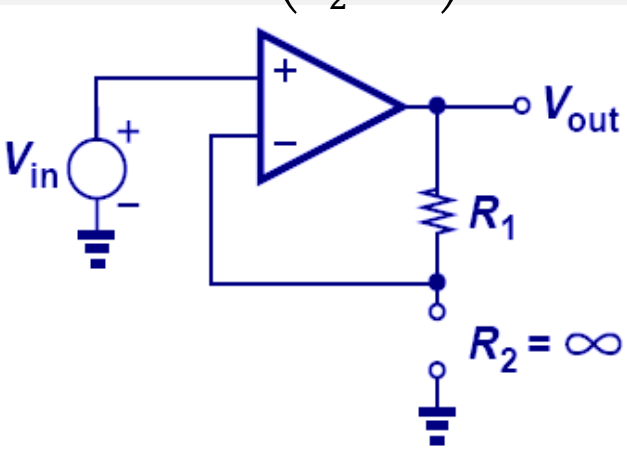
Extreme Cases of R_2 (Infinite A_0)

- If R_2 is zero, the loop is open and V_{out}/V_{in} is equal to the intrinsic gain of the op amp.
- If R_2 is infinite, the circuit becomes a unity-gain amplifier and V_{out}/V_{in} becomes equal to one.

$\left(\frac{R_1}{R_2} = \infty\right)$



$\left(\frac{R_1}{R_2} = 0\right)$

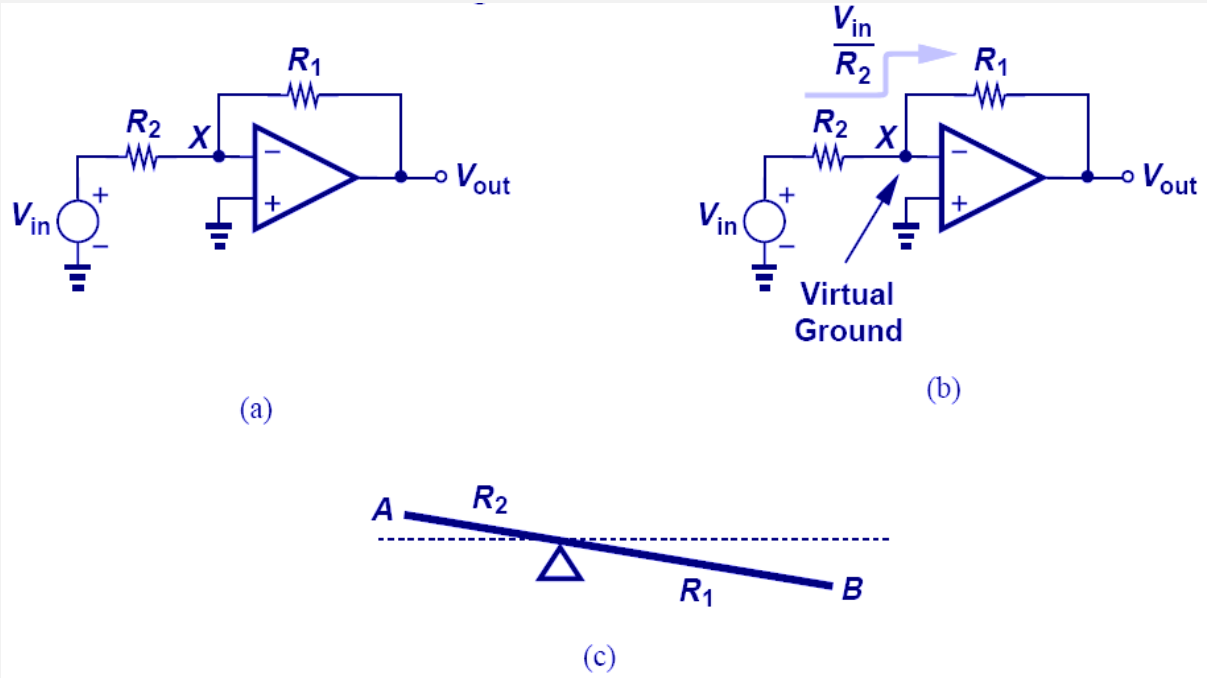


$$A_v = \left.\frac{A_0}{1 + \frac{R_2}{R_1 + R_2} A_0}\right|_{R_2=0} = A_0$$

$$A_v = \left.\frac{A_0}{1 + \frac{R_2}{R_1 + R_2} A_0}\right|_{R_2=\infty} = 1$$

Inverting Amplifier

- Infinite A_0 forces the negative input to be a virtual ground.



Based on the virtual short concept, $V_X = 0$ (virtual ground).

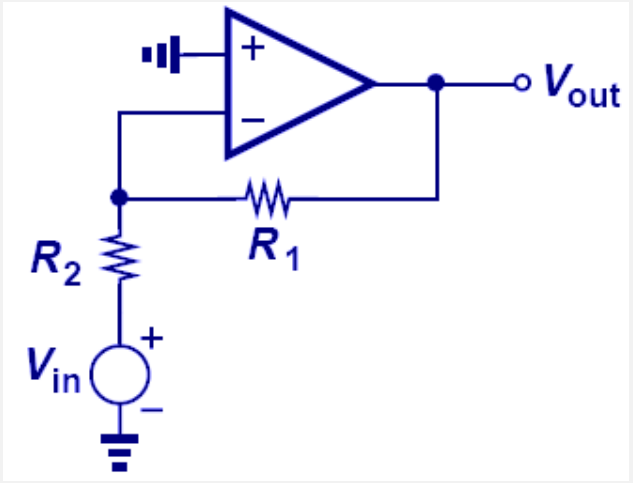
Apply the KCL at node X,

$$\frac{V_{in}}{R_2} = \frac{0 - V_{out}}{R_1}$$

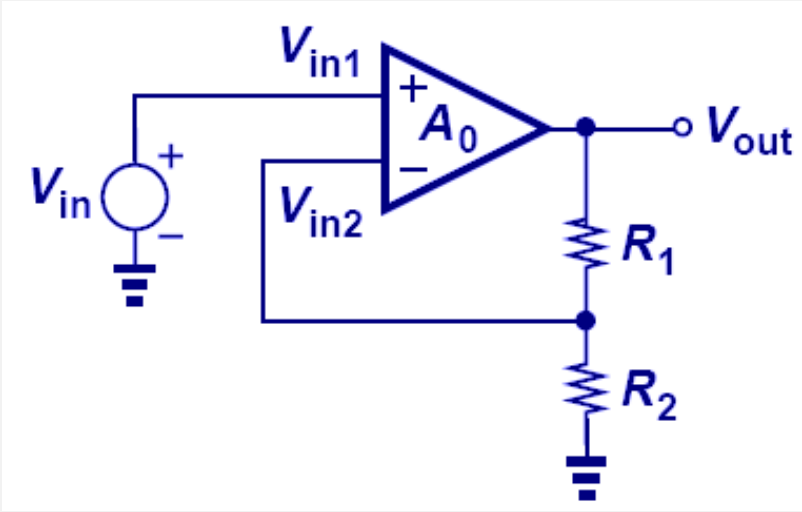
$$A_v = \frac{V_{out}}{V_{in}} = -\frac{R_1}{R_2}$$

Another View of Inverting Amplifier

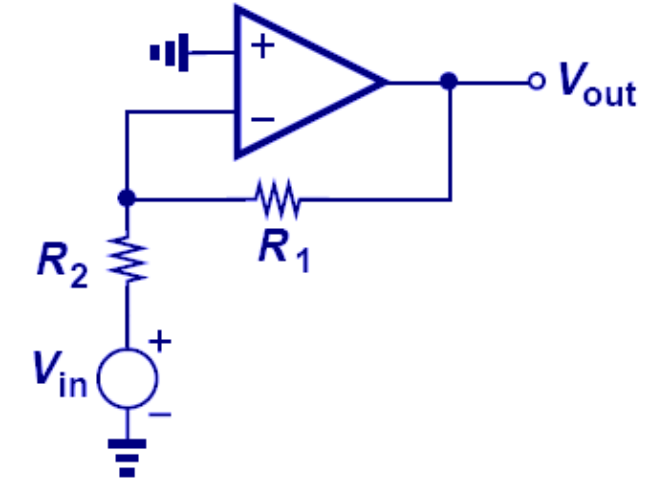
Inverting



Noninverting



Inverting Amplifier (Finite A_0)

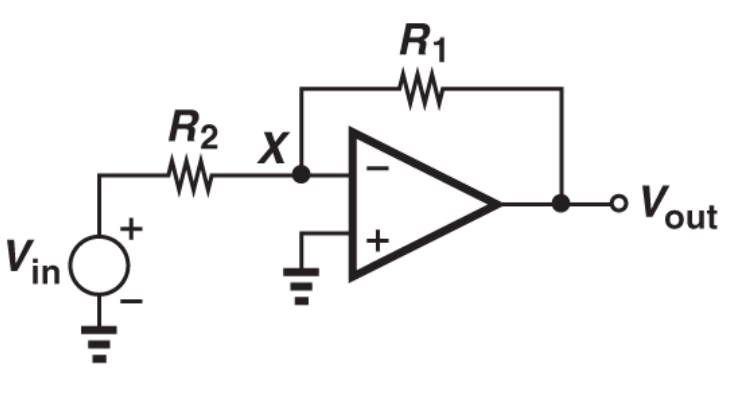


$$A_v = \frac{V_{out}}{V_{in}} \approx -\frac{R_1}{R_2}\left[1 - \left(1 + \frac{R_1}{R_2}\right)\frac{1}{A_0}\right]$$

where $\left(1 + \frac{R_1}{R_2}\right)\frac{1}{A_0}$ is the "gain error," it must be minimized according to each application's requirements. The larger the closed loop gain, the more inaccurate the circuit is.

Example 8-4

- Design the inverting amplifier for a nominal gain of 4, a gain error of 0.1%, and an input impedance of at least 10 kΩ.



Solution:

Given nominal gain and gain error ,

$$\frac{R_1}{R_2} = 4 \quad (1)$$

$$\left(1 + \frac{R_1}{R_2}\right)\frac{1}{A_0} = 0.1\% \quad (2)$$

Form (1) and (2),

$$(1+4)\frac{1}{A_0} = 0.1\%$$

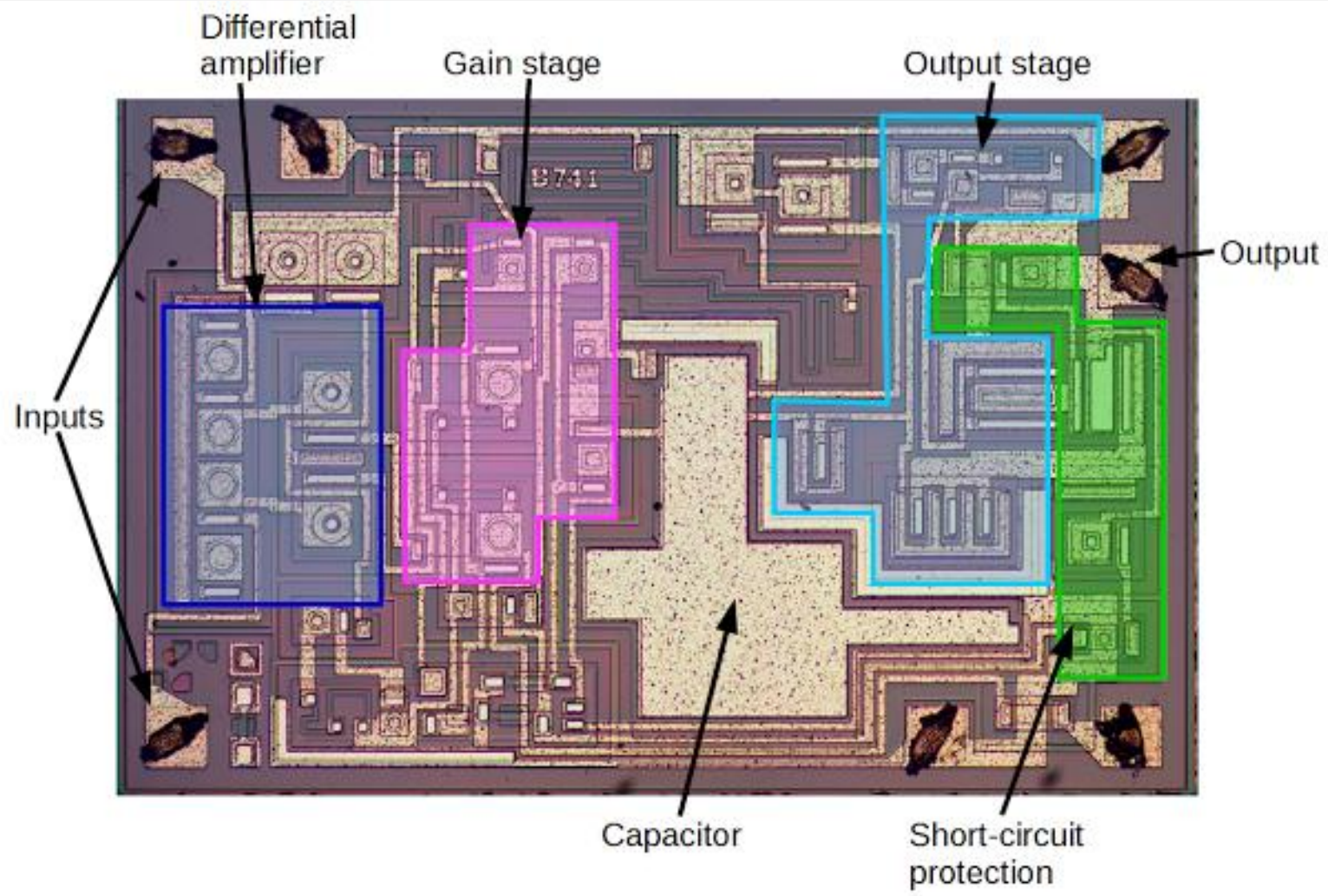
$$A_0 = 5000$$

Since the input impedance is approximately equal to R_2, we choose:

$$R_2 = 10\ \text{k}\Omega$$

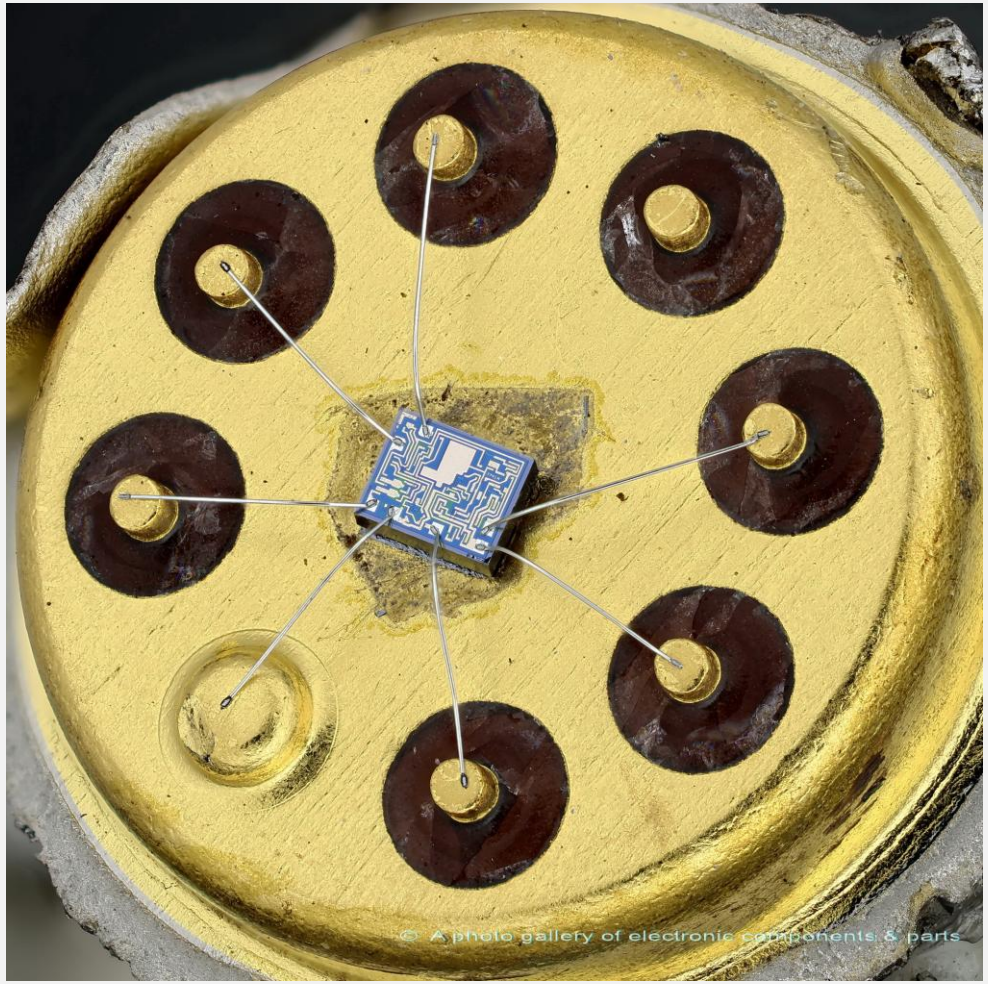
$$R_1 = 4R_2 = 40\ \text{k}\Omega$$

Miller compensation capacitor – The Big Cap

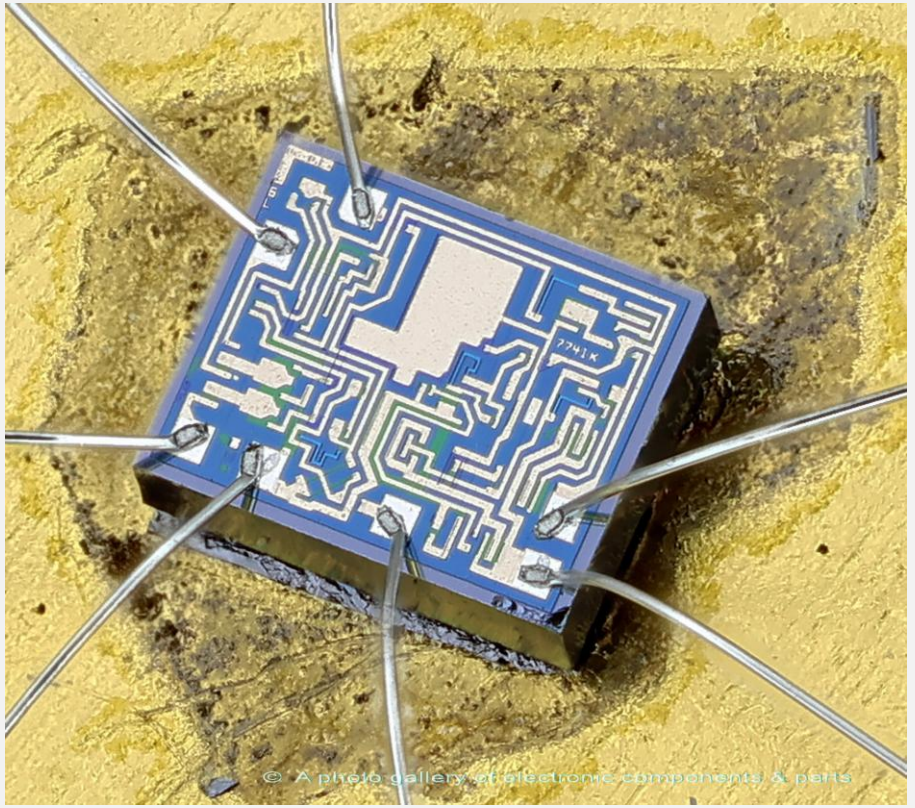


Ref: https://www.righto.com/2015/10/

Miller compensation capacitor – The Big Cap



μA741, Fairchild Semiconductor, 1978.
The device is short-circuit protected and the internal frequency compensation provides stability without external components.

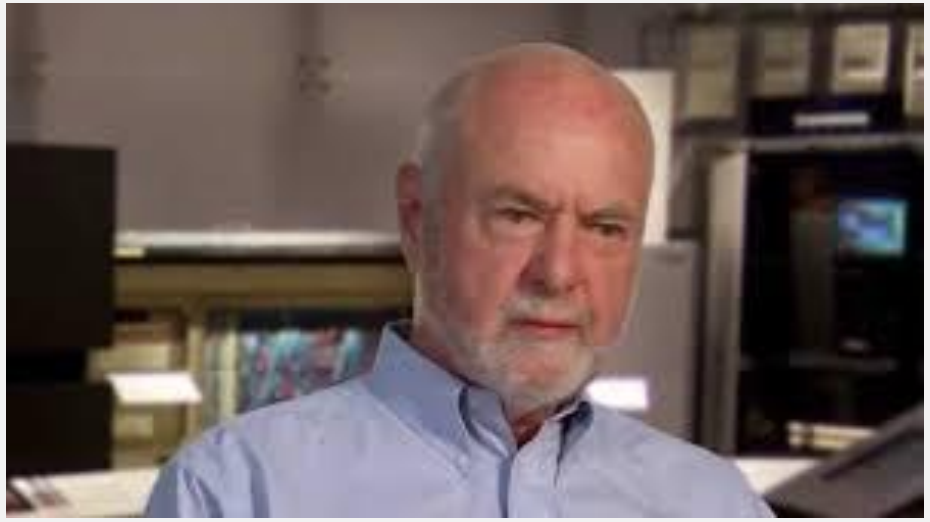




Ref: https://www.facebook.com/groups/259905524039336/search/?q=741

Dave Fullagar – the designer of μA741 at Fairchild

- The μA741 is an early operational amplifier (op amp) integrated circuit (IC), originally designed at Fairchild Semiconductor by Dave Fullagar and released in 1968.
- Dave Fullagar worked to improve the output dead zone, input stage impedance and included a compensation capacitor to improve the dynamic behaviour of the opamp.



μA733	Differential Video Amplifier	7-40
μA734	Precision Voltage Comparator	6-29
μA739	Dual Low-Noise Audio Preamplifier/Operational Amplifier	5-105
μA740	FET Input Operational Amplifier	5-109
μA741	Frequency Compensated Operational Amplifier	5-113
μA742	Zero Crossing AC Trigger-Trigac	7-46
μA747	Dual Frequency Compensated Operational Amplifier	5-122
μA748	Operational Amplifier	5-132
μA749	Dual Audio Operational Amplifier/Preamplifier	5-140

Ref: wiki